Table 1: Validity (%), joint MMD, and reference coverage (%) across five DGPs (three-seed means). Bold marks the best displayed method for each DGP and metric.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑	Valid. ↑	J-MMD ↓	Cover. ↑
Conditional diffusion (generation direction $Z \to X$)															
CD	13.55	0.1467	40.00	83.22	0.0025	78.52	78.12	0.0062	74.35	85.55	0.0034	70.49	80.65	0.0072	73.08
Compositional CD	75.50	0.1065	84.78	38.54	0.0139	31.25	31.74	0.0368	27.67	21.21	0.0497	16.06	20.93	0.1054	20.04
Extrapolative generation baselines															
Additive mean†	0.00	0.1174	0.00	**100.00**	0.0088	0.55	0.00	0.1711	0.00	0.00	0.0716	0.00	0.00	0.3416	0.00
Additive + residual†	0.18	0.2322	0.65	7.81	0.1407	6.17	0.00	0.2037	0.00	1.35	0.1523	0.86	0.18	0.3266	0.10
Fixed composition + residual†	0.13	0.2353	0.50	0.03	0.2327	0.00	0.02	0.2014	0.00	0.07	0.2548	0.00	0.00	0.3798	0.00
Rep4Ex-inspired + residual†	0.85	0.2714	2.87	3.68	0.2490	3.06	0.23	0.2732	0.02	0.39	0.2621	0.02	0.03	0.3648	0.03
Classifier-guided diffusion (likelihood direction $X_t \to Z$)															
CG: sparse factorized ($s = 1$)	46.93	0.1027	74.85	5.40	0.0776	1.84	8.92	0.1280	4.07	3.43	0.1096	0.75	4.07	0.2045	3.52
CG: dense factorized ($s = 1$)	55.43	0.1073	75.58	7.67	0.0692	2.86	3.45	0.1266	0.65	2.86	0.1073	0.80	1.69	0.2084	1.79
CG: dense joint ($s = 1$)	1.47	0.5295	4.27	15.35	0.0359	6.74	13.22	0.0745	5.49	14.75	0.0526	4.57	6.18	0.1360	4.82
CG: sparse factorized ($s = 2$)	76.13	0.0530	61.22	24.82	0.0161	7.42	35.74	0.0183	13.85	19.48	0.0383	4.87	36.04	0.0758	30.99
CG: dense factorized ($s = 2$)	**82.85**	0.0633	56.77	29.61	0.0116	9.83	26.16	0.0194	2.67	17.01	0.0423	1.14	5.70	0.0752	4.64
CG: dense joint ($s = 2$)	0.45	0.5508	1.42	18.20	0.0111	9.18	34.49	0.0143	13.77	25.72	0.0241	6.35	14.19	0.0394	9.52
SEDGE (ours)															
SEDGE: no weighting	12.00	0.2198	29.95	8.98	0.4006	20.82	8.97	0.3650	17.61	8.92	0.4248	17.12	2.46	0.6294	4.10
SEDGE-LGD ($s = 0.5$)	38.07	0.1525	70.00	44.11	0.1343	64.81	31.92	0.2048	49.61	47.90	0.1267	60.01	23.67	0.3420	35.40
SEDGE-LGD ($s = 1$)	56.80	0.0996	81.08	78.24	0.0104	79.17	60.17	0.0711	64.49	79.57	0.0147	70.05	66.37	0.1153	66.60
SEDGE-LGD ($s = 2$)	39.98	0.0272	57.12	77.07	0.0302	62.42	71.89	0.0302	50.91	79.46	0.0328	55.27	76.25	0.0516	56.77
SEDGE-LWS	77.55	**0.0225**	**89.97**	91.98	**0.0018**	**87.37**	**91.26**	**0.0024**	**84.29**	**91.55**	**0.0030**	**82.23**	**87.39**	**0.0016**	**78.86**